

Section 1: Basic SELECT Queries

1. Display all information about all employees

```
SELECT * FROM EMPLOYEE;
```

2. Employee number, name, and salary

```
SELECT EMPNO, ENAME, SAL FROM EMPLOYEE;
```

3. Unique job titles

```
SELECT DISTINCT JOB FROM EMPLOYEE;
```

4. Employee name and annual salary

```
SELECT ENAME, SAL * 12 AS ANNUAL_SALARY FROM EMPLOYEE;
```

5. Employees from department 10

```
SELECT * FROM EMPLOYEE WHERE DEPTNO = 10;
```

6. Name and job of CLERKs

```
SELECT ENAME, JOB FROM EMPLOYEE WHERE JOB = 'CLERK';
```

7. Names starting with 'S'

```
SELECT ENAME FROM EMPLOYEE WHERE ENAME LIKE 'S%';
```

8. Second letter is 'L'

```
SELECT ENAME FROM EMPLOYEE WHERE ENAME LIKE '_L%';
```

9. Name contains 'AR'

```
SELECT ENAME FROM EMPLOYEE WHERE ENAME LIKE '%AR%';
```

10. Hired in 1981

```
SELECT * FROM EMPLOYEE WHERE YEAR(HIREDATE) = 1981;
```

11. Salary more than 2500

```
SELECT ENAME, SAL FROM EMPLOYEE WHERE SAL > 2500;
```

12. Not MANAGER

```
SELECT * FROM EMPLOYEE WHERE JOB <> 'MANAGER';
```

13. Employees with commission

```
SELECT * FROM EMPLOYEE WHERE COMM IS NOT NULL;
```

14. Employees without manager

```
SELECT * FROM EMPLOYEE WHERE MGR IS NULL;
```

15. SALESMAN in department 30

```
SELECT ENAME FROM EMPLOYEE
```

```
WHERE DEPTNO = 30 AND JOB = 'SALESMAN';
```

16. CLERK or ANALYST

```
SELECT ENAME FROM EMPLOYEE
```

```
WHERE JOB IN ('CLERK', 'ANALYST');
```

17. Salary between 1000 and 2000

```
SELECT ENAME FROM EMPLOYEE
```

```
WHERE SAL BETWEEN 1000 AND 2000;
```

Section 2: Ordering and Grouping

18. Sort by name

```
SELECT * FROM EMPLOYEE ORDER BY ENAME;
```

19. Sort by department, salary descending

```
SELECT * FROM EMPLOYEE
```

```
ORDER BY DEPTNO, SAL DESC;
```

20. Total employees

```
SELECT COUNT(*) AS TOTAL_EMPLOYEES FROM EMPLOYEE;
```

21. Total salary

```
SELECT SUM(SAL) AS TOTAL_SALARY FROM EMPLOYEE;
```

22. Average salary

```
SELECT AVG(SAL) AS AVG_SALARY FROM EMPLOYEE;
```

23. Max and Min salary

```
SELECT MAX(SAL) AS MAX_SAL, MIN(SAL) AS MIN_SAL FROM EMPLOYEE;
```

24. Employees per department

```
SELECT DEPTNO, COUNT(*) AS EMP_COUNT  
FROM EMPLOYEE GROUP BY DEPTNO;
```

25. Average salary per job

```
SELECT JOB, AVG(SAL) AS AVG_SAL  
FROM EMPLOYEE GROUP BY JOB;
```

26. Total salary per department

```
SELECT DEPTNO, SUM(SAL) AS TOTAL_SAL  
FROM EMPLOYEE GROUP BY DEPTNO;
```

27. Departments with more than 3 employees

```
SELECT DEPTNO FROM EMPLOYEE  
GROUP BY DEPTNO HAVING COUNT(*) > 3;
```

28. Jobs with average salary > 2000

```
SELECT JOB FROM EMPLOYEE  
GROUP BY JOB HAVING AVG(SAL) > 2000;
```

Section 3: Joins and Subqueries

29. Employee name and department name

```
SELECT E.ENAME, D.DNAME  
FROM EMPLOYEE E  
JOIN DEPARTMENT D ON E.DEPTNO = D.DEPTNO;
```

30. Employee name, department, location

```
SELECT E.ENAME, D.DNAME, D.LOC  
FROM EMPLOYEE E  
JOIN DEPARTMENT D ON E.DEPTNO = D.DEPTNO;
```

31. Employee and manager name

```
SELECT E.ENAME AS EMPLOYEE, M.ENAME AS MANAGER  
FROM EMPLOYEE E
```

```
LEFT JOIN EMPLOYEE M ON E.MGR = M.EMPNO;
```

32. Employee name, salary, grade

```
SELECT E.ENAME, E.SAL, G.GRADE  
FROM EMPLOYEE E  
JOIN GRADE G ON E.SAL BETWEEN G.LOSAL AND G.HISAL;
```

33. Employees working in NEW YORK

```
SELECT E.ENAME  
FROM EMPLOYEE E  
JOIN DEPARTMENT D ON E.DEPTNO = D.DEPTNO  
WHERE D.LOC = 'NEW YORK';
```

34. Department where SMITH works

```
SELECT D.DNAME  
FROM EMPLOYEE E  
JOIN DEPARTMENT D ON E.DEPTNO = D.DEPTNO  
WHERE E.ENAME = 'SMITH';
```

35. Employees earning more than ALLEN

```
SELECT ENAME FROM EMPLOYEE  
WHERE SAL > (SELECT SAL FROM EMPLOYEE WHERE ENAME = 'ALLEN');
```

36. Same department as SCOTT

```
SELECT ENAME FROM EMPLOYEE  
WHERE DEPTNO = (SELECT DEPTNO FROM EMPLOYEE WHERE ENAME = 'SCOTT');
```

37. Same job as JONES

```
SELECT ENAME FROM EMPLOYEE  
WHERE JOB = (SELECT JOB FROM EMPLOYEE WHERE ENAME = 'JONES');
```

38. Managers managing more than 2 employees

```
SELECT MGR FROM EMPLOYEE  
GROUP BY MGR HAVING COUNT(*) > 2;
```

Section 4: Functions

39. Names in lowercase

```
SELECT LOWER(ENAME) FROM EMPLOYEE;
```

40. First three characters

```
SELECT SUBSTRING(ENAME, 1, 3) FROM EMPLOYEE;
```

41. Length of name

```
SELECT ENAME, LENGTH(ENAME) FROM EMPLOYEE;
```

42. Salary rounded to nearest thousand

```
SELECT ROUND(SAL, -3) FROM EMPLOYEE;
```

43. Remainder of salary ÷ 1000

```
SELECT MOD(SAL, 1000) FROM EMPLOYEE;
```

44. Current date and time

```
SELECT CURRENT_TIMESTAMP;
```

45. Hire year

```
SELECT ENAME, YEAR(HIREDATE) FROM EMPLOYEE;
```

46. Hire month name

```
SELECT ENAME, MONTHNAME(HIREDATE) FROM EMPLOYEE;
```

47. Years with company (2025)

```
SELECT ENAME, 2025 - YEAR(HIREDATE) AS YEARS_WORKED  
FROM EMPLOYEE;
```

Section 5: DDL and DML

48. Create DEPARTMENT table

```
CREATE TABLE DEPARTMENT (  
    DEPTNO INT PRIMARY KEY,  
    DNAME VARCHAR(30),
```

```
LOC VARCHAR(30)  
);
```

49. Add PINCODE column

```
ALTER TABLE DEPARTMENT ADD PINCODE INT;
```

50. Insert department 50

```
INSERT INTO DEPARTMENT VALUES (50, 'HR', 'HOUSTON');
```

51. Update SALES location

```
UPDATE DEPARTMENT
```

```
SET LOC = 'ATLANTA'
```

```
WHERE DNAME = 'SALES';
```

52. Delete OPERATIONS department

```
DELETE FROM DEPARTMENT WHERE DNAME = 'OPERATIONS';
```